Anbu

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Links

Overview: github.com/anbu-05 - landing page summarizing my work
Full Portfolio: github.com/anbu-05/Portfolio - detailed write-ups of projects and PCB designs

Skills and Education

Firmware: C, C++, Python, Embedded C, FreeRTOS, STM32 Cube HAL, ESP-IDF, MQTT, PID control
Digital Design: Verilog, SystemVerilog, RTL design, FPGA design, AXI4-Lite, RISC-V assembly
Hardware: STM32, ESP32, Siemens S7-1500 PLC, PCB design, schematic capture, layout, routing, SMD soldering
Tools: Git, GitHub, QuestaSim, Cadence Xcelium, Altium Designer, KiCad, RISC-V GCC, TIA Portal, S7-PLCSIM

Vellore Institute of Technology, Vellore | **B.Tech EE.VLSI** *Aug'23 – present | CGPA: 7.10*
Kamala Niketan Montessori School, Trichy | **Higher Secondary Education (12th)** *2023 | Board: 93.8%*

Projects | portfolio

Mars Rover Prototype | **PCB Designs** | *Embedded Systems, Wireless, Power Electronics, PCB, System Integration*

- **IRC 2025 (electrical junior):** designed a custom **LoRa** RF link (SX1278) for 1 km bidirectional telemetry - we chose this instead of off-the-shelf RC to carry mixed data like arm position, science feedback, encoder states, on a noise-resistant channel
- Wrote firmware across 4 **ESP32** nodes using a shared data stream on a single UART channel
- **IRC 2026 (Electrical Head):** Designed and iteratively built the rover PCB suite (images) including:
  - **200A / 24V / 4.8 kW, 6 rail** power distribution board with stackable motor driver slots and per-rail current sensing
  - motor drivers using half H-Bridge ICs, to be easily repairable
  - **STM32F103/STM32H523** development boards tailored per node (LoRa, drive train, robotic arm, environmental sensors)
- Designed breakout boards adapting commercial devboards to rover pinouts for rover-specific nodes; all boards implement protection circuitry

PicoRV32-based RISC-V SoC | *RTL, SoC Design, FPGA, Verilog/SystemVerilog*

- Built a RISC-V SoC around **PicoRV32** with memory-mapped SRAM, UART, and GPIO
- Designed a custom **AXI4-Lite interconnect** with address offsetting - each peripheral assumes `0x0`, keeping peripheral RTL fully decoupled from the system map so that new peripherals plug in without touching the core
- Built a signature-based verification flow - firmware writes pass/fail values to reserved memory, eliminating the need to change the testbench every time we add a new peripheral
- Wrote C firmware using the **riscv-gcc toolkit** and **Makefile** build system. Currently migrating to **Cadence Xcelium** for a unified toolchain toward physical design

Experience

Team Vyadh | *Electrical Head* *July 2025 – July 2026*

- Led a 6-member electrical domain within a 33-member rover team - set delivery timelines, managed budget and inventory, and mentored 3 juniors to where they can independently handle most technical work
- Sourced components across 5+ vendors, balancing specs, availability, and cost - instead of defaulting to the cheapest
- Migrated all rover nodes to STM32 and implemented CAN across 6 nodes for noise immunity and hot-swap capability in a high-EMI drivetrain
- Defined domain-wide PCB standards - mandatory protection circuitry and standardized board dimensions for panel sharing to reduce fabrication cost

Antropi Robotics, Bengaluru | *Electronics & Firmware Intern* *May 2026 – July 2026*

- Designed and deployed **ESP32-S3** conveyor-control firmware using **ESP-IDF**, **FreeRTOS**, and **MQTT** - built a job-based state machine handling tray sensor logic, timeouts, and status feedback across 4 conveyor nodes
- Designed and fabricated 2 iterations of the **conveyor controller PCB**, revising the sensor interface and board-level fixes after a sensor change
- Built reusable **PID**-based motor-control firmware and a **PID**-tuning GUI, commissioning position control for an autodoor system

Vanderlande India Pvt. Ltd., Noida | *PLC Controls Intern* *May 2025 – June 2025*

- Programmed PLCs and built HMI interfaces for baggage handling systems using **Siemens S7-1500** and **TIA Portal**
- Designed a PLC-controlled conveyor system with modular Function Blocks and validated real-time control logic using FactoryIO and **S7-PLCSIM** (GitHub Repository)